

REDOX BIOCHEMISTRY OF CELLULAR THIOLS

Abstract

The redox state of cellular thiols plays an imperative role in maintaining homeostasis by regulating signal transduction and protein behaviour. S-glutathionylation, a reversible post-translational oxidative modification, which results in derivatization of specific target protein cysteine residues with glutathione has often been found to be associated with certain disease pathologies particularly neurodegeneration and cancer. Reduced glutathione (GSH) is the most prevalent cellular antioxidant, present in 5mM-10mM physiological concentration and the imbalance in the GSH/GSSG ratio as a result of S-glutathionylation is linked to impaired cellular signalling, apoptosis and protein misfolding. However, in certain cases, S-glutathionylation can also rescue the redox-active sulfhydryl residues in a crucial target protein from overoxidation, thus preventing permanent target site inactivation. For quantitative analysis of cellular thiols, DTNB (5,5'-dithio-bis (2-nitrobenzoic acid)) assay is predominantly performed. This colorimetric assay is based on a thiol disulfide exchange reaction, followed by spectrophotometric analysis of the strongly colored product, TNB, with an absorption maximum at 412nm. The DTNB assay can also be standardized to detect S-glutathionylated sulfhydryl residues in a target protein sample of interest. In contrast to a simple DTNB assay, GSH is added to the reduced protein sample in a novel Reverse DTNB procedure. The future prospects of Reverse DTNB assay, when accompanied by Mass spectrometry can provide a better understanding of the active thiols susceptible to S-glutathionylation thus facilitating the analysis of protein function during cellular stress conditions.

Keywords: Reduced glutathione (GSH); S-glutathionylation; cellular thiols

1. Introduction

The two major classes of chemically reactive molecules containing free radicals of oxygen and nitrogen are referred to as the reactive oxygen and nitrogen species (ROS and RNS) respectively. Under normal physiological conditions, both ROS and RNS are produced as by-products in redox reactions to maintain redox homeostasis. The production of different ROS is achieved by the reduction step of oxygen (O_2) to water (H_2O) which includes superoxide ($O_2^{\bullet-}$), hydrogen peroxides (H_2O_2), hydroxyl ions (OH^-) and nitric oxide (NO) [1]. The major cellular sites to produce ROS include mitochondria, endoplasmic reticulum, peroxisomes, and NADPH oxidase (NOX). The major defence mechanisms used by the immune cells (macrophages and neutrophils) against foreign invasion is the production of ROS/RNS that damages target cell lipids, proteins and DNA [2, 3]. Beyond the role of a weapon, ROS and RNS are found to act as messenger molecules for redox signalling. Proteins with cysteine thiols ($-SH$) undergo post-translational modifications (PTMs) that trigger ROS/RNS-mediated signal transduction regulation redox homeostasis.

Glutathione (GSH) is a low molecular weight cellular tripeptide (L--glutamate-L-cysteinyl-glycine) serving as the most prevalent cellular antioxidant. Reduced glutathione (GSH) is present in only millimolar concentrations (5mM-10mM) yet is as crucial as water and oxygen in the body. Like any other antioxidants operating in the body, GSH also serves as an important intracellular defence against oxidative stress. The ratio of oxidized glutathione which is Glutathione Disulfide (GSSG) to the reduced form of Glutathione (GSH) acts as a biomarker for the level of intracellular oxidative stress and helps in maintaining redox balance. The ratio of reduced glutathione to oxidized glutathione (GSH: GSSG) acts as a biomarker for diagnosing many diseases or disorders as it is a direct indicator of oxidative stress. The diagnosis of cancer, which is one of the most concerning diseases, can also be achieved by analysing this ratio [4]. Thus, GSH and the ratio of GSH: GSSG plays significant roles in medical science, and it can be used soon to enhance the rate of diagnosis and treatment of various diseases.[Figure 1]

2. Materials and Methods

2.1. Preparation of Protein Sample

To guarantee experimental precision and dependability, a bovine serum albumin (BSA) stock solution was rigorously manufactured. A concentration of 5 mg/ml was methodically obtained to ensure the best circumstances for future investigations and tests [5]. To improve precision in experimental design, BSA concentrations were changed to micromolar (M) units for easier comparison and interpretation of data. The extraction and preparation of cellular thiols from cultivated cells followed strict methods to assure methodological consistency and the integrity of future studies. Cultured cells were meticulously washed three times with phosphate-buffered saline (PBS) to remove any extraneous pollutants and culture medium residues. Centrifugation of the cell suspension was carried out from 600 to 700 revolutions per minute (rpm) for 5 minutes and the pellet obtained was refrigerated at -20°C. Prior to lysis, the stored pellet was carefully resuspended in PBS. Following resuspension, the pellet was subjected to a series of freeze-thaw cycles, each lasting 10 minutes and alternating between -20°C and ambient temperature (25°C). After completing the freeze-thaw cycles, the lysate was centrifuged at 10,000 rpm for 10 minutes. The centrifugation resulted in supernatant rich in cellular thiols which was saved for further studies.

2.2. Fractionation and reverse DTNB Assay Execution

The fractionation and assay execution procedure began with the removal of excess Dithiothreitol (DTT) from the protein sample. Using a 30 kDa cut-off filter, centrifugation at 4000 rpm for 30 minutes at 4°C aided in the separation of excess DTT, which precipitated into the holder, while the protein containing free thiols remained within the cut-off filter. To ensure volume integrity, the separated excess DTT was methodically transferred to a separate microcentrifuge tube (MCT), with the cut-off filter receiving a compensating injection of phosphate buffer with EDTA (PB+EDTA). Iterative centrifugation and DTT collection cycles were performed until the disulfide bonds were completely reduced, as demonstrated by the lack of DTT in the holder. The collected DTT in the MCT was then treated with 5,5'-dithiobis-(2-nitrobenzoic acid) (DTNB), which caused a reaction with free thiols to produce S-TNB. Before continuing, we confirmed the holder's absence

from DTT. [6] In parallel, DTNB solution was introduced to the protein sample that remained within the cut-off filter, allowing it to react with free thiols and produce S-TNB. A further centrifugation stage at 4000 rpm for 30 minutes at 4°C enabled the collection of excess, unreactive TNB into the holding tube. After confirming full DTT removal and quantifying excess TNB, glutathione (GSH) was added to the protein sample, causing glutathionylation reactions with cysteine residues to produce S-glutathionylated proteins (SSG). The resultant S-TNB transformed to SSG was kept within the cut-off filter for further analysis.

2.3. Spectrophotometric Analysis

Following the separation and assay execution steps, spectrophotometric analysis was used to determine the level of glutathionylation and the number of thiol groups present in the protein sample. Thiol quantification entailed measuring the sample's absorbance at 412 nm, a wavelength chosen for its effectiveness in detecting thiol groups and simplifying thiol concentration determination. Additionally, absorbance measurements at 412 nm and 700 nm were taken to test glutathionylation. The absorbance at 412 nm revealed the existence and quantity of thiols, while the absorbance at 700 nm served as a reference to guarantee an accurate evaluation by considering background absorbance.

2.4. Bradford Assay

To quantify the total protein content of the protein sample, the Bradford test, a popular protein quantification method, was used. For this test, the spectrophotometer was set to photometry mode with a wavelength of 575 nm. Initially, the absorbance of the Coomassie dye was zeroed, and then the protein sample and the Bradford reagent were added, and the absorbance at the appropriate wavelength was measured.

3. Results

The results obtained from the Reverse DTNB assay were as follows:

1. In case of reduced BSA sample, three biological replicative experiments revealed that one, three and two thiols of BSA sample are susceptible to S-Glutathionylation respectively.[Figure 2]
2. In case of cell lysate sample, the assay with KB cell lysate sample yielded non- significant result, whereas two replicative experiments with NMuMG cell lysate sample showed 75.72 M and 53.91M PSSGs/mg/ml thiols of total protein concentration which are susceptible to S-Glutathionylation respectively.[Figure 3]

3.1. Figures, Tables and Schemes

All figures and tables are cited in the main text, e.g. Figure 1, Figure 2, Figure 3, and Table 1.

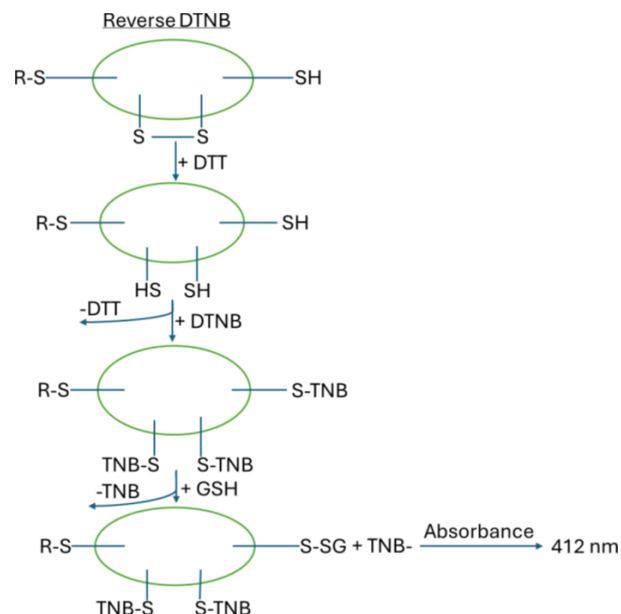


Figure 1: Flowchart diagram depicting the procedure for reverse DTNB assay.

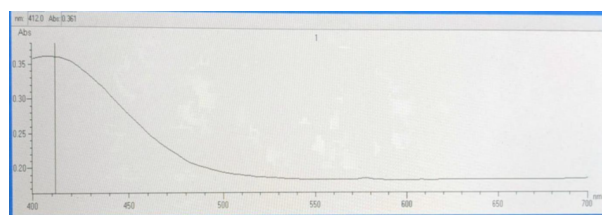


Figure 2: The absorbance curve for BSA Sample 2 indicating baseline absorbance(700nm) value as 0.185 and peak absorbance(412nm) value as 0.361.

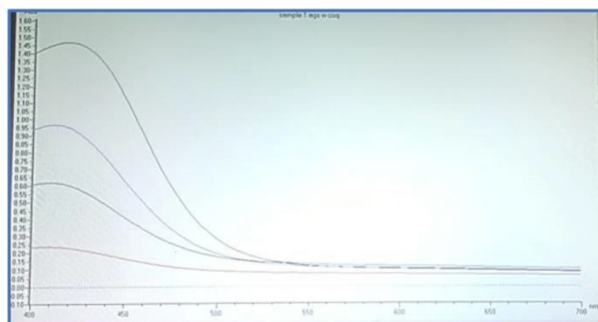


Figure 3: The absorbance curve for NMuMG cell lysate. (The first line from the top represents no dilution. The second line and the third line from the top represents 4X dilution and 2X dilution (respectively)).

S.No.	Sample	Initial Concentration	Total number of thiols	Total S-glutathionylated thiol conc. per TNB ₂ -release	Protein estimation by Bradford assay	Thiols susceptible to S-glutathionylation
1	BSA	5 mg/ml (75.18 μ M)	35	41.4 μ M	0.3 mg/ml	3
2	BSA	5 mg/ml (75.18 μ M)	35	35.7 μ M	1.2 mg/ml	2
3	NM _u MG cell lysate	200 μ l	Unidentified (a complex mix of proteins)	37.86 μ M	0.5 mg/ml	75.72 μ M PSSGs/mg/ml of total protein conc.
4	NM _u MG cell lysate	100 μ l	Unidentified (a complex mix of proteins)	12.13 μ M	0.225 mg/ml	53.91 μ M PSSGs/mg/ml of total protein conc.

Table 1: Calculation table for number of thiols susceptible to S-glutathionylation.

4. Discussion

After analysing the experimental results, it can be concluded that, approximately 2.85%, 8.57% and 5.71% thiols of reduced BSA sample were susceptible to S-Glutathionylation. This observed percentage of glutathionylated thiols significantly reflects the specificity of such post-translational modification occurring particularly at specific thiol residues in BSA. The experimental results obtained in case of cell lysate sample did not provide exact percentages of S-Glutathionylated thiols as a complex mixture of proteins are present in the sample. The reasons contributing to such non-significant experimental result can be multiple, namely, oxidation of active cysteinyl residues in the proteins to a greater extent, errors in sample handling such as inaccurate pipetting, inaccurate freezing, and thawing cycles. It might also result from the inaccessibility of thiols to Glutathione due to solvent exposure owing to continuous spins after DTT treatment. Another contributing factor can be inefficient removal of reagents which might possibly cause interference with other reagents added in the process.

References

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